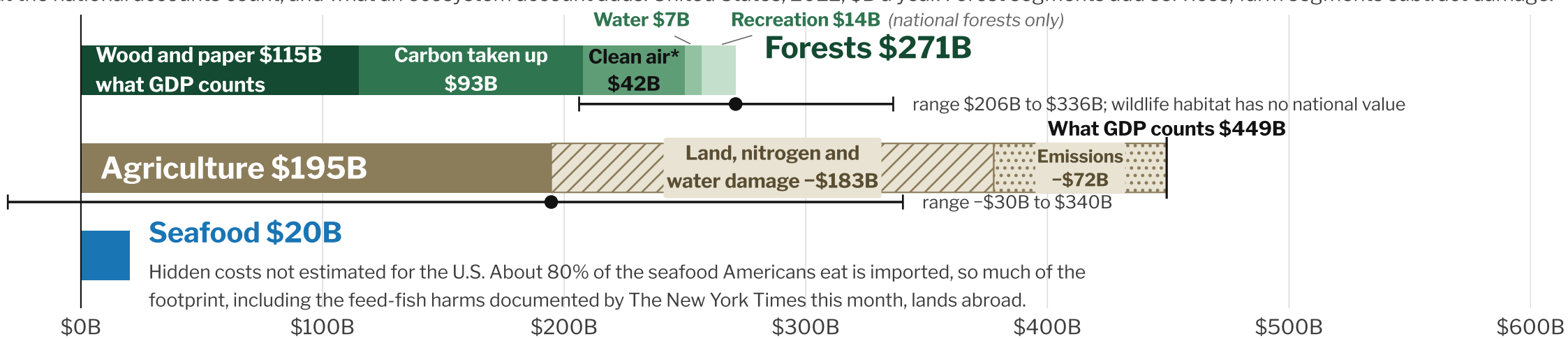
U.S. forests give back roughly twice what GDP counts, and wood not grown here is cut somewhere else

GDP credits forests with about $115B a year in wood and paper. Carbon, clean air, water and recreation add $90B to $220B, so GDP captures about a third to a half of what forests provide. The carbon sink is weakening as forests age and disturbances mount, harvest given up at home is mostly made up by cutting elsewhere, and forest research receives a seventh of the federal capacity funds that farm research receives.

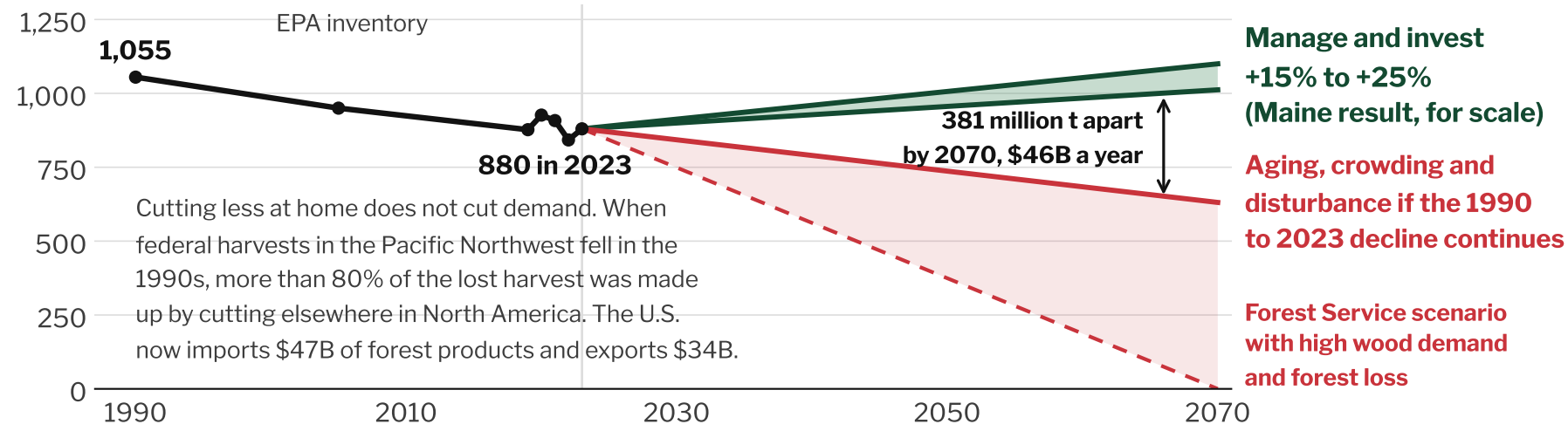
1. Undervalued

What the national accounts count, and what an ecosystem account adds. United States, 2022, $B a year. Forest segments add services; farm segments subtract damage.



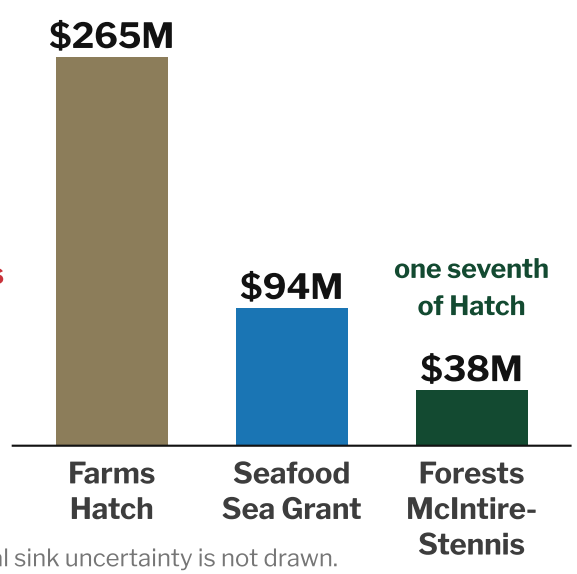
2. Misunderstood

Net carbon uptake by U.S. forests, million t CO_2e a year. Paths after 2023 are illustrations, not forecasts



3. Underinvested

Federal research capacity funds, FY2024



Notes: Ranges are scenario ranges, not statistical confidence intervals; the national accounts publish no sampling error for value added, and EPA's annual sink uncertainty is not drawn. Panel 1: value added less depreciation (BEA 2022). Forest carbon is the 2022 net uptake (772 million t, EPA) at $51 to $190 per t (central $120); clean air from Cavender-Bares et al. (2022); water (Sedell et al. 2000, inflated) and recreation (USDA Forest Service) cover national forests only; *clean air covers all U.S. trees, including urban trees; forest carbon is net of fire and other disturbance emissions, but wildfire smoke damage is not charged; dollar years and methods are mixed and the values are welfare measures, so the total is indicative and not comparable with Bhutan's account. Agriculture is charged its own emissions and FAO (2023) estimates of land, nitrogen and water damage (range −57% to +100%). Seafood is BEA marine living resources; import share from NOAA Fisheries of the United States 2023. Panel 2: EPA inventory 1990 to 2023, forest carbon stock change including wood products, years EPA tabulates. Paths are illustrations built from published end points, not forecasts: manage and invest applies the 15% to 25% carbon gain from mixed management in Maine (Daigneault et al. 2024) to the 2023 national sink; the decline path extends the 1990 to 2023 change; the net source scenario is from the 2020 RPA Assessment (high wood demand, forest loss); EPA attributes the 1990 to 2023 decline to aging forests and rising fire and other disturbance. Leakage: Wear and Murray (2004), via Murray (2019); trade from UN Comtrade, HS 44, 47 and 48, 2024. The 2070 gap prices the distance between the manage and invest low edge and the trend path at $120 per t. Panel 3: Hatch, McIntire-Stennis, Sea Grant core plus aquaculture, FY2024 enacted. Chart: Weiskittel, Daigneault and Woodall, University of Maine.